

Table 6. Number of detected *cis*-eQTLs in transcript-wise analysis of three harmonized RNA NGS datasets.

	Number of unique genes with <i>cis</i> -eQTLs	
	Without principal component correction	With principal component correction
Montgomery <i>et al.</i> (paired-end RNA-seq)	94	145
Pickrell <i>et al.</i> (single-end RNA-seq)	199	438
NTR-NESDA transcript-wise (DeepSAGE)	292	579
Meta-analysis	651	1,207

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and environmental effects. As such, compensating for these non-genetic effects would increase the power to detect *cis*-eQTL effects. To mitigate the effects of non-genetic sources of variability, we first $\log_2$ transformed the data and centered and scaled each tag, and subsequently applied PCA on the sample correlation matrix. We then used the first PCs as covariates, and re-did the non-parametric *cis*-eQTL mapping on the residual expression data (using the procedure described by [6]).

Validation of genotype-dependent alternative polyadenylation in RNA-seq datasets

The genomic coordinates of the 3'-UTR, obtained from Refseq Genes, were split into two separate regions (distal and proximal 3'-UTRs) according to the position of the DeepSAGE tags with opposite directions, the position of LongSAGE tags from CGAP, and the

position of reported and predicted polyadenylation sites from polyA_DB database. To calculate the coverage in proximal and distal regions in RNA-seq datasets, we created a coverage histogram from each .bam alignment file using coverageBed tool from BEDTools package (version 2.17.0) [39]. Subsequently, a custom Python script was used to convert the histogram in number of nucleotides mapped per region, normalized by the length of the region. The ratio between the number of counts in the proximal region and the distal region was then calculated.

qPCR validation of alternative polyadenylation

Expression of short and long variants of *HPS1* and *IRF5* was quantified by qRT-PCR, which was performed on a subset of RNA samples used for the DeepSAGE sequencing. cDNA was synthesized from 400 ng of total RNA using BioScript MMLV

Table 7. Trait-associated SNPs detected in the sequencing-based transcript-wise meta-analysis, but not detected in array-based eQTL dataset of 1,469 peripheral blood samples.

SNP name	Chr.	Transcript position (midpoint)	<i>Cis</i> -regulated gene	Associated trait
rs1052501	3	41963564	<i>ULK4</i>	Multiple myeloma
rs347685	3	141782879	<i>TFDP2</i>	Chronic kidney disease
rs4580814	5	1081324	<i>SLC12A7</i>	Hematological and biochemical traits
rs4947339	6	28911984	<i>C6orf100</i>	Platelet aggregation
rs2517532	6	31024818	<i>HCG22</i>	Hypothyroidism
rs2844665	6	31024818	<i>HCG22</i>	Stevens-Johnson syndrome and toxic epidermal necrolysis (SJS-TEN)
rs6457327	6	31024818	<i>HCG22</i>	Follicular lymphoma
rs3130501	6	31324124	<i>HLA-B</i>	Stevens-Johnson syndrome and toxic epidermal necrolysis (SJS-TEN)
rs2858870	6	32434437	<i>HLA-DRB9</i>	Nodular sclerosis Hodgkin lymphoma
rs3129889	6	32434437	<i>HLA-DRB9</i>	Multiple sclerosis
rs3135388	6	32434437	<i>HLA-DRB9</i>	Multiple sclerosis
rs477515	6	32434437	<i>HLA-DRB9</i>	Inflammatory bowel disease
rs9271100	6	32524134	<i>HLA-DRB6</i>	Systemic lupus erythematosus
rs9273349	6	32632106	<i>HLA-DQB1</i>	Asthma
rs3807989	7	116183034	<i>CAV1</i>	PR interval
rs12680655	8	135604552	<i>ZFAT</i>	Height
rs4929923	11	8642408	<i>TRIM66</i>	Menarche (age at onset)
rs12785878	11	71161461	<i>RP11-660L16.2</i>	Vitamin D insufficiency
rs12580100	12	56436876	<i>RPS26</i>	Psoriasis
rs4924410	15	40329664	<i>SRP14</i>	Ewing sarcoma
rs7364180	22	42184613	<i>MEI1</i>	Alzheimer's disease biomarkers

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